

Lab 3: Architecture Justification

Disaster Relief Supply & Volunteer Coordinator | Yele Charan | PES1UG25CS856

Architecture Selection: We chose **Layered Architecture** for the **Disaster Relief Supply & Volunteer Coordinator System**.

Architectural Choice

The system is organised into three layers. The **Presentation Layer** holds the Coordination Dashboard, a web UI with role-based views for the Disaster Manager, Volunteers and Warehouse Staff. The **Business Layer** holds the Access Control Gateway and three services: Supply Management, Volunteer Coordination and Delivery Tracking. The **Data Layer** holds the Disaster Relief Database. Each layer talks only to the layer directly below it, through defined provided/required interfaces (see the component diagram).

Reason 1: Separation of responsibilities matches the system's functions

The requirements split into clearly different jobs: matching and prioritising supplies (FR-001, FR-003), assigning volunteers by skill (FR-002), tracking vehicles and deliveries (FR-004) and giving the Disaster Manager one overview (FR-005). In a layered design each job is its own business component behind an interface, and the dashboard only displays results. A change such as new volunteer-skill rules touches one component and does not affect the UI or the database. Where jobs overlap, as when a dispatch is created from allocated supplies in UC-03, Delivery Tracking simply uses Supply Management.

Reason 2: One consistent data store, and a simple structure to deploy quickly

Allocations, volunteer assignments and the delivery manifest all read and write the same warehouse inventory and shelter-requirement data. The inventory check in UC-03 must always see the latest stock, or two requests could be given the same supplies. A central database behind a business layer keeps this data consistent. A disaster-response team also needs something small enough to set up and operate quickly, and layered architecture avoids the operational overhead of running many independent services.

Security Advantage (NFR-002)

The Presentation Layer has no path to the database. Every request must pass through the **Access Control Gateway** in the Business Layer, which authenticates the user and applies role-based permissions (Disaster Manager, Volunteer, Warehouse Staff) *before* any service runs. Unauthenticated or unauthorised requests are rejected at this single point, so sensitive shelter, volunteer and logistics data is never exposed directly to the UI. One entry point is also easier to audit than many.

Performance Benefit (NFR-001)

Disaster areas often have weak or unstable networks. The Business Layer sits next to the data, so it can filter, prioritise and combine records (for example, the FR-005 overview) into one compact JSON response containing only the fields the dashboard needs. The dashboard makes few, small requests instead of downloading raw records or making many round trips. Calls between the gateway, services and database are internal to one deployment, so they add no extra network hops over the user's slow link.

Styles considered and rejected

Style	Why it was not chosen
Microservices	Independent scaling is not needed for this system. It would split the shared inventory data and add network latency and operational complexity that suit neither a small team nor a low-bandwidth environment.
Client-Server	A single server with no internal structure would mix UI logic, rules and data access. It would also make security checks and later changes harder to isolate.

Trade-off accepted: each extra layer adds a small processing cost and the business layer is a shared dependency. For a coordination system of this size, that cost is far smaller than the benefits above.